



What is an echo?

When you make a sound, invisible sound waves spread out in all directions. If the sound waves meet a hard surface like a cave ceiling or wall, they bounce back, and some sound reaches your ears. What you hear is a copy of the sound you made, only fainter.

1 Original shout

When you shout, you make a vibration in your throat. This vibration travels away from your mouth quickly as a sound wave. A sound wave is a wave of squashed and stretched air, shown here in red.

Quick quiz



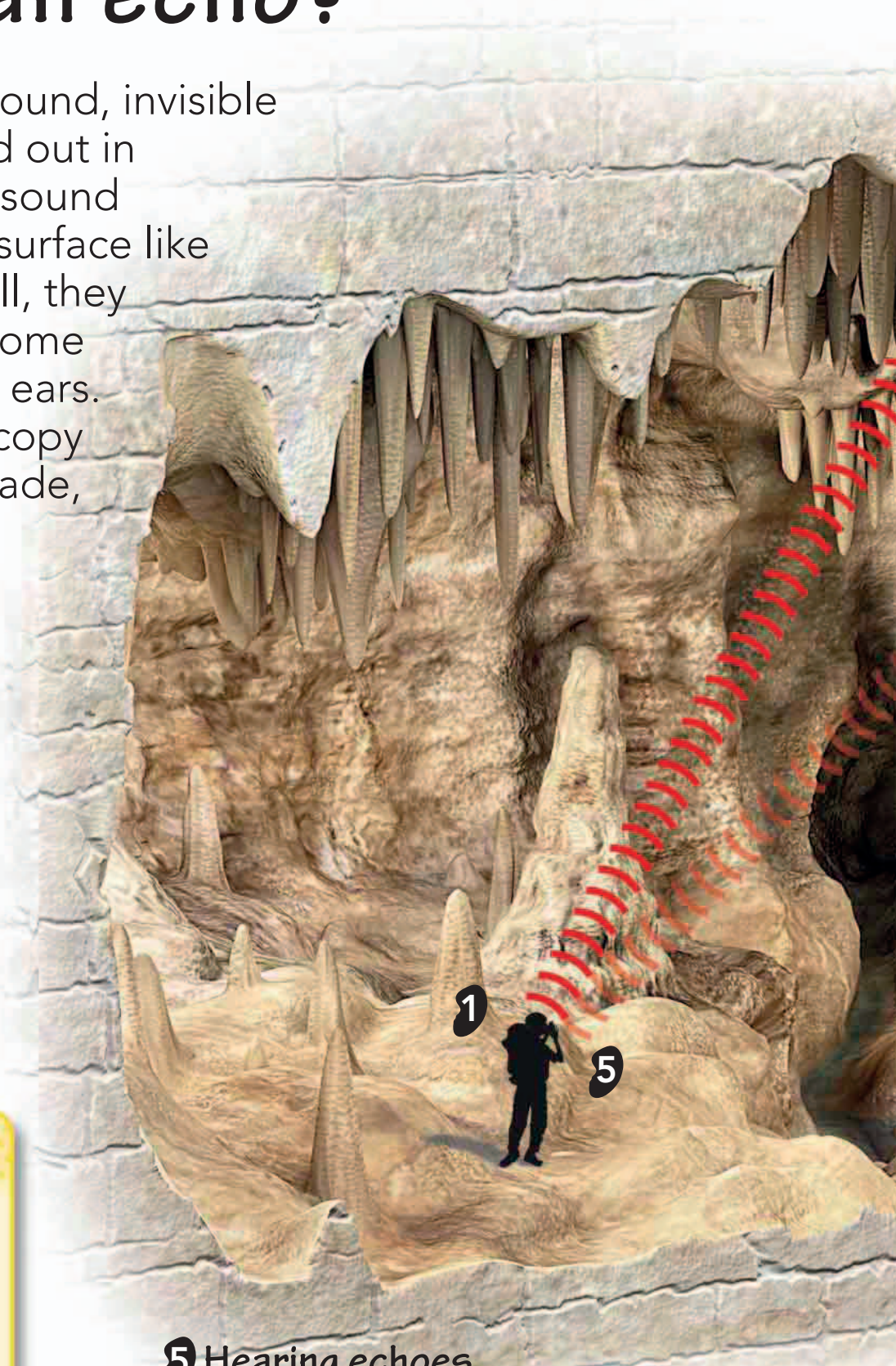
How does sound travel through air?



Why are echoes loud and clear in a cave?



Which is louder, a baby or a jet engine?



5 Hearing echoes

When the sound waves reach your ears, you hear the echoes of your shout. In a cave, echoes are clear and loud because the rocky surfaces reflect sound well.